

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

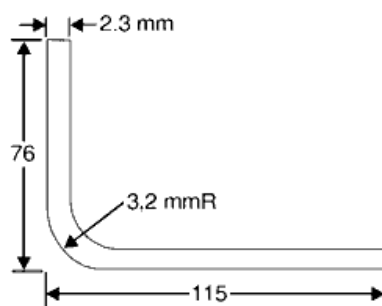
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|---|-------|---|
| A) State & explain the advantages of manufacturing the products from Sheet Metal. | (CO2) | 6 |
| B) Enlist any six important properties of sheet metal material and explain in short, any two of them. | (CO2) | 6 |
| C) Write short note on any two of the following commonly used sheet metal materials
a) Stainless Steel b) Aluminum c) Copper | (CO2) | 6 |

Q.2 Solve Any Two of the following.

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|--|-------|---|
| A) Explain the difference between punching and blanking operation with neat sketches and suitable examples. | (CO3) | 6 |
| B) Explain the mechanics of shearing with neat sketches. | (CO3) | 6 |
| C) Estimate the punching force required to punch the hole of 20mm diameter in a plate having the thickness of 2 mm. The maximum shear strength of the material is 400 MPa. | (CO3) | 6 |

Q. 3 Solve Any Two of the following.

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|---|-------|---|
| A) Estimate the length of sheet necessary to manufacture the component as shown in the fig.1. The thickness of the sheet is 2.3mm and it is made of C10 material. | (CO3) | 6 |
|---|-------|---|



(All Dimensions are in mm)

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|----|---|-------|---|
| B) | Write short note on any two of the following. a) Deep Drawing Operation
b) Ironing Operation c) Defects in Drawn parts | (CO3) | 6 |
| C) | Explain with neat sketches, V-Bending and Wipe Bending along with suitable examples. | (CO3) | 6 |

Q.4 Solve Any Two of the following.

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|----|---|-------|---|
| A) | Explain working of progressive Die giving suitable example with neat sketch | (C04) | 6 |
| B) | Explain working of Compound Die giving suitable example with neat sketch. | (CO4) | 6 |
| C) | Classify and explain types of presses with their applications. | (CO5) | 6 |

Q. 5 Solve Any Two of the following.

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|----|--|-------|---|
| A) | Describe any one case study of sheet metal manufacturing highlighting the following.
a) Generic name of the product with uses b) Material Used c) Types of Sheet Metal operations involved d) Sequence of Operations with brief description. | (CO1) | 6 |
| B) | State the importance of center of pressure and how it is determined? Explain with suitable example. | (CO1) | 6 |
| C) | A washer having the outside diameter of 30mm and inside diameter of 20mm is to be made. Determine the die and punch sizes required for punch & dies used in punching and blanking operations. Thickness of the sheet is 2.5mm and shear strength of the material is 450 MPa. | (CO1) | 6 |

***** End *****